2012 Team Answers

T-1. $64 - 64\pi + 32\pi\sqrt{2}$

T-2. 81

T-3. 165

T-4. 820

T-5. $10 + 2\sqrt{2}$

T-6. $-\dfrac{23}{60}$

T-7. $\dfrac{9}{5}$

T-8. 47

T-9. 6

T-10. $\dfrac{189\sqrt{11}}{11}$

2012 Team Solutions

T-1. Because $AB = 16$, $AC = BC = \frac{16}{\sqrt{2}} = 8\sqrt{2}$. Then each of the large arcs has radius 8, and the small arc has radius $8\sqrt{2} - 8$. Each large arc has measure $45°$ and the small arc has measure $90°$. Therefore the area enclosed by each large arc is $\frac{45}{360} \cdot \pi \cdot 8^2 = 8\pi$, and the area enclosed by the small arc is $\frac{90}{360} \cdot \pi \cdot \left(8\sqrt{2} - 8\right)^2 = 48\pi - 32\pi\sqrt{2}$. Thus the sum of the areas enclosed by the three arcs is $64\pi - 32\pi\sqrt{2}$. On the other hand, the area of the triangle is $\frac{1}{2}\left(8\sqrt{2}\right)^2 = 64$. So the area of the desired region is $\mathbf{64 - 64\pi + 32\pi\sqrt{2}}$.

T-2. Begin by partitioning $\{2, 3, \ldots, 50\}$ into the subsets

$$\begin{aligned} A &= \{2, 4, 8, 16, 32\} \\ B &= \{3, 9, 27\} \\ C &= \{5, 25\} \\ D &= \{6, 36\} \\ E &= \{7, 49\} \\ F &= \text{all other integers between 2 and 50, inclusive.} \end{aligned}$$

If $\log_b a$ is rational, then either a and b are both members of one of the sets A, B, C, D, or E, or $a = b \in F$ (see note below for proof). Then the number of possible ordered pairs is

$$\begin{aligned} |A|^2 + |B|^2 + |C|^2 + |D|^2 + |E|^2 + |F| &= 25 + 9 + 4 + 4 + 4 + 35 \\ &= \mathbf{81}. \end{aligned}$$

Note: The solution essentially relies on one fact: if $\log_b a$ is rational, then b and a are both integral powers of some integer d. This fact can be demonstrated as follows. Suppose that $\log_b a = m/n$; then $b^{m/n} = a \Rightarrow (b^{1/n})^m = a$. Let $d = b^{1/n}$. The proof that d is an integer requires a significant theorem in number theory: if m and n are relatively prime positive integers, there exist integers x and y such that $mx - ny = 1$. Now consider the set $S = \left\{\frac{a^x}{b^y} \mid x, y \in \mathbb{Z}\right\}$. Because a and b are rational and x and y are integers, the elements $\frac{a^x}{b^y}$ are all rational numbers. But because $a = d^m$ and $b = d^n$, the expression $\frac{a^x}{b^y} = \frac{d^{mx}}{d^{ny}} = d^{mx-ny}$. By the theorem mentioned previously, there exist integers x and y such that $mx - ny = 1$, and $d^{mx-ny} \in S$. So d^1 is rational, hence d is rational. Because the nth root of an integer is either an integer or an irrational number, d must be an integer.

T-3. Condition on the number n of A's that appear in the word; n is at least two, because of the requirement that A occur more often than any other letter, and n is at most 4, because of the requirement that there be at least two distinct letters. In the case $n = 4$, there are 3 choices for the other letter, and 5 choices for where to place it, for a total of 15 possibilities. In the case $n = 3$, there are two possibilities to consider: either a second letter occurs twice, or there are two distinct letters besides A. If a second letter occurs twice, there are 3 choices

for the other letter, and $\frac{5!}{3!\cdot 2!} = 10$ ways to arrange the three A's and two non-A's, for their locations, for a total of 30 choices. If there are two distinct letters besides A, then there are $\binom{3}{2} = 3$ ways to pick the two letters, and $\frac{5!}{3!\cdot 1!\cdot 1!} = 20$ ways to arrange them, for a total of 60 words. Thus there are a combined total of 90 words when $n = 3$. In the case $n = 2$, no other letter can occur twice, so all the letters R, M, L, must appear in the word; they can be arranged in $\frac{5!}{2!\cdot 1!\cdot 1!\cdot 1!} = 60$ ways. The total number of words satisfying the conditions is therefore $15 + 90 + 60 = \mathbf{165}$.

T-4. Let d be the common difference of the sequence. Then $a_{a_2} = a_1 + (a_2 - 1)d = 100 \Rightarrow (a_2 - 1)d = 90$. But $a_2 = a_1 + d = 10 + d$, so $(9 + d)d = 90$. Solving the quadratic yields $d = -15$ or $d = 6$, but the requirement that a_i be positive for all i rules out the negative value, so $d = 6$ and $a_n = 10 + (n - 1) \cdot 6$. Thus $a_3 = 10 + 2(6) = 22$, and $a_{a_3} = a_{22} = 10 + 21(6) = 136$. Finally, $a_{a_{a_3}} = a_{136} = 10 + 135(6) = \mathbf{820}$.

T-5. First, note that both graphs are symmetric about the y-axis, so C and D must be reflections of B and A, respectively, across the y-axis. Thus $x_C = -x_B$ and $y_C = y_B$, so $BC = 2x_C$. For $x < 0$, the equations become $y = x^2 + x - 12$ and $y = -x - k$; setting the x-expressions equal to each other yields the equation $x^2 + 2x + (k - 12) = 0$, from which $x = -1 \pm \sqrt{13 - k}$. Therefore $x_B = -1 + \sqrt{13 - k}$ and $BC = 2 - 2\sqrt{13 - k}$. (Note that the existence of two distinct negative values of $-1 \pm \sqrt{13 - k}$ forces $12 < k \leq 13$.)

Thus the x-coordinates of the four points are

$$\begin{aligned} x_A &= -1 - \sqrt{13 - k} \\ x_B &= -1 + \sqrt{13 - k} \\ x_C &= 1 - \sqrt{13 - k} \\ x_D &= 1 + \sqrt{13 - k}. \end{aligned}$$

To compute y_A, use the second equation $y = |x| - k$ to obtain $y_A = 1 + \sqrt{13 - k} - k = (1 - k) + \sqrt{13 - k}$; similarly, $y_B = (1 - k) - \sqrt{13 - k}$. Therefore

$$\begin{aligned} AB &= \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2} \\ &= \sqrt{(2\sqrt{13 - k})^2 + (-2\sqrt{13 - k})^2} \\ &= 2\sqrt{2(13 - k)}. \end{aligned}$$

Because $AB = BC$, $2\sqrt{2(13 - k)} = 2 - 2\sqrt{13 - k}$. Let $u = \sqrt{13 - k}$; then $2\sqrt{2}\,u = 2 - 2u$, from which $u = \frac{2}{2 + 2\sqrt{2}} = \frac{1}{1 + \sqrt{2}}$, which equals $\sqrt{2} - 1$ by rationalizing the denominator. Thus

$$13 - k = (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}, \text{ so } k = \mathbf{10 + 2\sqrt{2}}.$$

Because $10 + 2\sqrt{2} \approx 12.8$, the value of k determined algebraically satisfies the inequality $12 < k \leq 13$ observed above.

Alternate Solution: Let $C = (a, b)$. Because C and D lie on a line with slope 1, $D = (a + h, b + h)$ for some $h > 0$. Because both graphs are symmetric about the y-axis, the other two points of intersection are $A = (-a - h, b + h)$ and $B = (-a, b)$, and $a > 0$.

In terms of these coordinates, the distances are $AB = CD = \sqrt{2}\,h$ and $BC = 2a$. Thus the condition $AB = BC = CD$ holds if and only if $\sqrt{2}\,h = 2a$, or $h = \sqrt{2}\,a$.

The foregoing uses the condition that C and D lie on a line of slope 1, so now use the remaining equation and subtract:

$$\begin{aligned} b &= a^2 - a - 12; \\ b + h &= (a + h)^2 - (a + h) - 12; \\ h &= 2ah + h^2 - h. \end{aligned}$$

Because the points are distinct, $h \neq 0$. Dividing by h yields $2 - 2a = h = \sqrt{2}\,a$. Thus $a = \frac{2}{2+\sqrt{2}} = 2 - \sqrt{2}$.

Finally, because C lies on the two graphs, $b = a^2 - a - 12 = -8 - 3\sqrt{2}$ and $k = a - b = \mathbf{10 + 2\sqrt{2}}$.

T-6. There are $6! = 720$ permutations of the zeros, so the average value is the sum, S, divided by 720. Setting any particular zero as A leaves $5! = 120$ ways to permute the other five zeros, so over the 720 permutations, each zero occupies the A position 120 times. Similarly, fixing any ordered pair (B, C) of zeros allows $4! = 24$ permutations of the other four zeros, and $BC = CB$ means that each *value* of BC occurs 48 times. Finally, fixing any ordered triple (D, E, F) allows $3! = 6$ permutations of the other variables, and there are $3! = 6$ equivalent arrangements within each product DEF, so that the product of any three zeros occurs 36 times within the sum. Let $S_1 = A + B + C + D + E + F$ (i.e., the sum of the zeros taken singly), $S_2 = AB + AC + \cdots + AF + BC + \cdots + EF$ (i.e., the sum of the zeros taken two at a time), and $S_3 = ABC + ABD + \cdots + DEF$ be the sum of the zeros three at a time. Then $S = 120S_1 + 48S_2 + 36S_3$. Using the sums and products of roots formulas, $S_1 = -2/1 = -2$, $S_2 = 3/1 = 3$, and $S_3 = -5/1 = -5$. Thus $S = 120(-2) + 48(3) + 36(-5) = -276$. The average value is thus $-\dfrac{276}{720} = -\dfrac{\mathbf{23}}{\mathbf{60}}$.

T-7. Compute the desired area as $[EGFB] - [EHB]$. To compute the area of concave quadrilateral $EGFB$, draw segment $\overline{BG}$, which divides the quadrilateral into three triangles, $\triangle DEG, \triangle BDG$, and $\triangle BGF$. Then $[BDG] = [DEG] = 18$ because the triangles have equal bases and heights. Because D, G, and F are collinear, to compute $[BGF]$ it suffices to find the ratio DG/GF. Use Menelaus's Theorem on $\triangle ADF$ with Menelaus Line $\overline{EC}$ to obtain

$$\frac{AE}{ED} \cdot \frac{DG}{GF} \cdot \frac{FC}{CA} = 1.$$

Because E and F are the midpoints of $\overline{AD}$ and $\overline{CA}$ respectively, $AE/ED = 1$ and $FC/CA = 1/2$. Therefore $DG/GF = 2/1$, and $[BGF] = \frac{1}{2}[BDG] = 9$. Thus $[EGFB] = 18+18+9 = 45$.

To compute $[EHB]$, consider that its base $\overline{EB}$ is twice the base of $\triangle DEG$. The ratio of their heights equals the ratio EH/EG because the altitudes from H and G to $\overleftrightarrow{BE}$ are parallel to each other. Use Menelaus's Theorem twice more on $\triangle AEC$ to find these values:

$$\frac{AD}{DE}\cdot\frac{EG}{GC}\cdot\frac{CF}{FA} = 1 \Rightarrow \frac{EG}{GC} = \frac{1}{2} \Rightarrow EG = \frac{1}{3}EC, \text{ and}$$
$$\frac{AB}{BE}\cdot\frac{EH}{HC}\cdot\frac{CF}{FA} = 1 \Rightarrow \frac{EH}{HC} = \frac{2}{3} \Rightarrow EH = \frac{2}{5}EC.$$

Therefore $\frac{EH}{EG} = \frac{2/5}{1/3} = \frac{6}{5}$. Thus $[EHB] = \frac{6}{5}\cdot 2\cdot[DEG] = \frac{216}{5}$. Thus $[FGH] = 45 - \frac{216}{5} = \mathbf{\frac{9}{5}}$.

Alternate Solution: The method of mass points leads to the same results as Menelaus's Theorem, but corresponds to the physical intuition that masses on opposite sides of a fulcrum balance if and only if the products of the masses and their distances from the fulcrum are equal (in physics-speak, the net torque is zero). If a mass of weight 1 is placed at vertex B and masses of weight 2 are placed at vertices A and C, then $\triangle ABC$ balances on the line $\overleftrightarrow{BF}$ and also on the line $\overleftrightarrow{CE}$. Thus it balances on the point H where these two lines intersect. Replacing the masses at A and C with a single mass of weight 4 at their center of mass F, the triangle still balances at H. Thus $BH/HF = 4$.

Next, consider $\triangle BEF$. Placing masses of weight 1 at the vertices B and E and a mass of weight 4 at F, the triangle balances at G. A similar argument shows that $DG/GF = 2$ and that $EG/GH = 5$. Because $\triangle DEG$ and $\triangle FHG$ have congruent (vertical) angles at G, it follows that $[DEG]/[FHG] = (DG/FG)\cdot(EG/HG) = 2\cdot 5 = 10$. Thus $[FGH] = [DEG]/10 = \dfrac{18}{10} = \mathbf{\dfrac{9}{5}}$.

T-8. Let $\alpha = 3+\sqrt{5}$ and $\beta = 3-\sqrt{5}$, so that $N = \lfloor \alpha^{34} \rfloor$, and let $M = \alpha^{34} + \beta^{34}$. When the binomials in M are expanded, terms in which $\sqrt{5}$ is raised to an odd power have opposite signs, and so cancel each other out. Therefore M is an integer. Because $0 < \beta < 1$, $0 < \beta^{34} < 1$, and so $M - 1 < \alpha^{34} < M$. Therefore $M - 1 = N$. Note that α and β are the roots of $x^2 = 6x - 4$. Therefore $\alpha^{n+2} = 6\alpha^{n+1} - 4\alpha^n$ and $\beta^{n+2} = 6\beta^{n+1} - 4\beta^n$. Hence $\alpha^{n+2} + \beta^{n+2} = 6(\alpha^{n+1} + \beta^{n+1}) - 4(\alpha^n + \beta^n)$. Thus the sequence of numbers $\{\alpha^n + \beta^n\}$ satisfies the recurrence relation $c_{n+2} = 6c_{n+1} - 4c_n$. All members of the sequence are determined by the initial values c_0 and c_1, which can be computed by substituting 0 and 1 for n in the expression $\alpha^n + \beta^n$, yielding $c_0 = (3+\sqrt{5})^0 + (3-\sqrt{5})^0 = 2$, and $c_1 = (3+\sqrt{5})^1 + (3-\sqrt{5})^1 = 6$. Then

$$\begin{aligned} c_2 &= (3+\sqrt{5})^2 + (3-\sqrt{5})^2 = 6c_1 - 4c_0 = 36 - 8 = 28, \\ c_3 &= (3+\sqrt{5})^3 + (3-\sqrt{5})^3 = 6c_2 - 4c_1 = 168 - 24 = 144, \end{aligned}$$

and because the final result is only needed modulo 100, proceed using only remainders modulo 100.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$c_n \bmod 100$	6	28	44	52	36	8	4	92	36	48	44	72	56	48	64	92	96

n	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$c_n \bmod 100$	8	64	52	56	28	44	52	36	8	4	92	36	48	44	72	56	48

Thus N leaves a remainder of $48 - 1 = \mathbf{47}$ when divided by 100.

Alternate Solution: As in the previous solution, let $\alpha = 3 + \sqrt{5}$ and $\beta = 3 - \sqrt{5}$, so that $N = \alpha^{34} + \beta^{34} - 1$ as argued above.

A straightforward way to compute powers of α and β is by successive squaring. Paying attention to just the last two digits of the integer parts yields the following values:

$$\begin{aligned}
\alpha^2 &= 14 + 6\sqrt{5};\\
\alpha^4 &= 196 + 180 + 168\sqrt{5} \equiv 76 + 68\sqrt{5};\\
\alpha^8 &\equiv 96 + 36\sqrt{5};\\
\alpha^{16} &\equiv 96 + 12\sqrt{5};\\
\alpha^{32} &\equiv 36 + 4\sqrt{5};\\
\alpha^{34} = \alpha^2 \cdot \alpha^{32} &\equiv 24 + 72\sqrt{5}.
\end{aligned}$$

Similarly, replacing $\sqrt{5}$ with $-\sqrt{5}$ yields $\beta^{34} \equiv 24 - 72\sqrt{5}$. Thus

$$N \equiv \left(24 + 72\sqrt{5}\right) + \left(24 - 72\sqrt{5}\right) - 1 \equiv \mathbf{47} \pmod{100}.$$

Alternate Solution: As in the previous solutions, let $\alpha = 3 + \sqrt{5}$ and $\beta = 3 - \sqrt{5}$, so that $N = \alpha^{34} + \beta^{34} - 1$ as argued above.

Now consider the binomial expansions more carefully:

$$\begin{aligned}
\alpha^{34} &= 3^{34} + \binom{34}{1}3^{33}\sqrt{5} + \binom{34}{2}3^{32}\cdot 5 + \binom{34}{3}3^{31}\cdot 5\sqrt{5} + \cdots + \binom{34}{33}3\cdot 5^{16}\sqrt{5} + 5^{17};\\
\beta^{34} &= 3^{34} - \binom{34}{1}3^{33}\sqrt{5} + \binom{34}{2}3^{32}\cdot 5 - \binom{34}{3}3^{31}\cdot 5\sqrt{5} + \cdots - \binom{34}{33}3\cdot 5^{16}\sqrt{5} + 5^{17};\\
N &= 2\left(3^{34} + \binom{34}{2}3^{32}\cdot 5 + \cdots + \binom{34}{32}3^2\cdot 5^{16} + 5^{17}\right) - 1.
\end{aligned}$$

The following argument shows that every term that is summarized by the ellipsis $(\cdots)$ in the expression for N is a multiple of 50. First, each such term has the form $\binom{34}{2k}3^{34-2k}5^k$, where $2 \le k \le 15$.

Thus it is enough to show that the binomial coefficient is even. Because $\binom{34}{2k} = \binom{34}{34-2k}$, it is enough to check this for $2 \le k \le 8$. Keep track of powers of 2: $\binom{34}{2}$ is an integer, so

$\binom{34}{4} = \binom{34}{2} \cdot \frac{32 \cdot 31}{3 \cdot 4}$ is a multiple of 2^3; $\binom{34}{6} = \binom{34}{4} \cdot \frac{30 \cdot 29}{5 \cdot 6}$ is also a multiple of 2^3; $\binom{34}{8} = \binom{34}{6} \cdot \frac{28 \cdot 27}{7 \cdot 8}$ is a multiple of 2^2; and so on.

It can also be shown that the sum of the last two terms is a multiple of 50. Again, there are plenty of factors of 5, so it is enough to note that both terms are odd, because $\binom{34}{32} = \frac{34 \cdot 33}{1 \cdot 2} = 17 \cdot 33$.

Thanks to the initial factor of 2 in the expression for N (outside the parentheses), the previous paragraphs show that $N \equiv 2\big(3^{34} + \binom{34}{2} 3^{32} \cdot 5\big) - 1 \pmod{100}$.

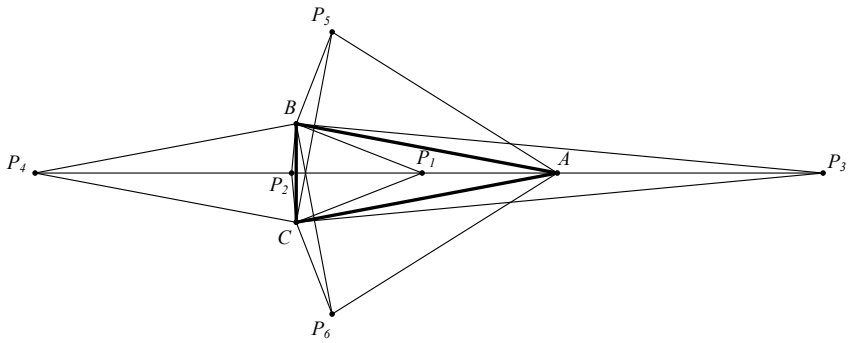
Now consider the powers of 3. Because $3^4 = 81$, we find that $3^8 = 80^2 + 2 \cdot 80 + 1 \equiv 61 \pmod{100}$, $3^{12} \equiv 41 \pmod{100}$, $3^{16} \equiv 21 \pmod{100}$, and $3^{20} \equiv 1 \pmod{100}$. (Note: those familiar with Euler's generalization of Fermat's Little Theorem will recognize this as an example, because $\phi(25) = 25 - 5 = 20$.) Therefore $3^{32} = 3^{20} \cdot 3^{12} \equiv 41 \pmod{100}$ and $3^{34} = 3^2 \cdot 3^{32} \equiv 69 \pmod{100}$.

Finally, $N \equiv 2\big(69 + 17 \cdot 33 \cdot 41 \cdot 5\big) - 1 \equiv 2 \cdot 69 + 10 \cdot \big(17 \cdot 33 \cdot 41\big) - 1 \equiv 38 + 10 - 1 \equiv \mathbf{47} \pmod{100}$.

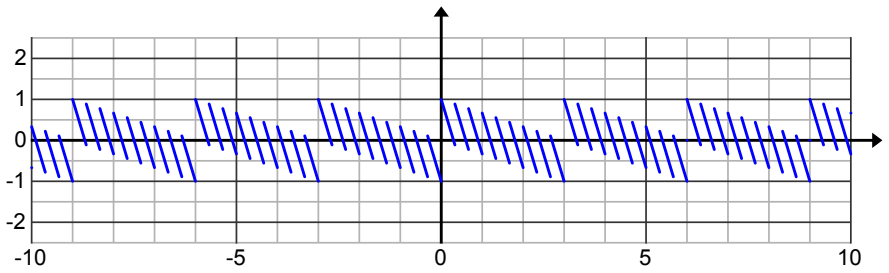
T-9. Focus on $\triangle PBC$. Either $PB = PC$ or $PB = BC$ or $PC = BC$.

If $PB = PC$, then P lies on the perpendicular bisector l of side $\overline{BC}$. Considering now $\triangle PAB$, if $PA = PB$, then $PA = PC$, and P must be the circumcenter of $\triangle ABC$; call this location P_1. If $PA = AB$, then $PA = AC$, and P, B, C all lie on a circle with center A and radius AB. There are two intersection points of that circle with l, one on each arc with endpoints B and C; label the one on the major arc P_2 and on the minor arc P_3. Finally, if $PB = AB$, then $PB = AC$ by the transitive property and $PC = AC$ by the perpendicular bisector theorem, so $PBAC$ is a rhombus; P is the reflection of A across $\overline{BC}$. Call this point P_4.

If $PB = BC$, then P must lie on the circle centered at B with radius BC. Considering $\triangle PAB$, if $PA = AB$, then P lies on the circle centered at A with radius AB. Now $\odot A$ and $\odot B$ intersect at two points, but one of them is C, so the other intersection must be the location of P, which is P_5. The condition $PB = AB$ is impossible, because it implies that $AB = BC$, which is false because in $\triangle ABC$, $\mathrm{m}\angle C > \mathrm{m}\angle A = 20^\circ$, so $AB > BC$. The third possibility for $\triangle PAB$ is that $PA = PB$, implying that the perpendicular bisector of $\overline{AB}$ intersects $\odot B$, which only occurs if $BC/AB \geq 1/2$ (although if $BC/AB = 1/2$, the triangle is degenerate). But $BC/AB = 2\cos 80^\circ$, and the given approximation $\cos 80^\circ \approx 0.17$ implies that $BC/AB \approx 0.34$. Hence the perpendicular bisector of $\overline{AB}$ does not intersect $\odot B$. Thus the assumption $PB = BC$ yields only one additional location for P, P_5. Similarly, $PC = BC$ yields exactly one more location, P_6, for a total of **6** points. All six points, and their associated triangles, are pictured below.



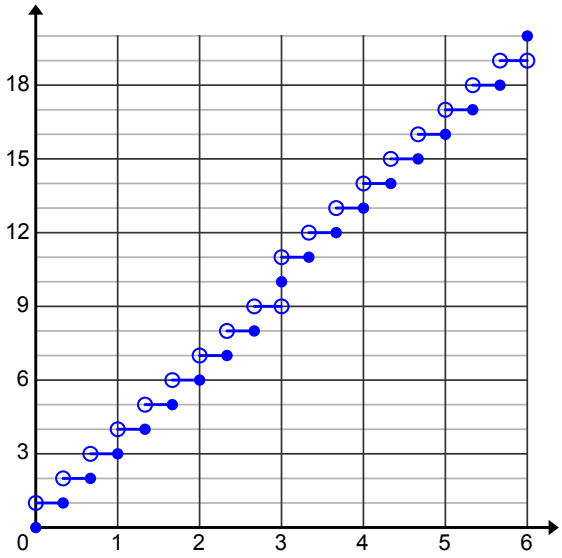
T-10. Let $f(x) = \lfloor \frac{x}{3} \rfloor + \lceil 3x \rceil$. Observe that $f(x+3) = f(x)+1+9 = f(x)+10$. Let $g(x) = f(x) - \frac{10}{3}x$. Then g is periodic, because $g(x+3) = f(x) + 10 - \frac{10x}{3} - \frac{10 \cdot 3}{3} = g(x)$. The graph of g is shown below:



Because $g(x)$ is the (vertical) distance between the graph of $y = f(x)$ and the line $y = \frac{10}{3}x$, the fact that g is periodic implies that f always stays within some fixed distance D of the line $y = \frac{10}{3}x$. On the other hand, because $\frac{10}{3} > \sqrt{11}$, the graph of $y = \frac{10}{3}x$ gets further and further away from the graph of $y = \sqrt{11}x$ as x increases. Because the graph of $y = f(x)$ remains near $y = \frac{10}{3}x$, the graph of $y = f(x)$ drifts upward from the line $y = \sqrt{11}x$.

For each integer n, define the open interval $I_n = (\frac{n-1}{3}, \frac{n}{3})$. In fact, f is constant on I_n, as the following argument shows. For $x \in I_n$, $\frac{n}{9} - \frac{1}{9} < \frac{x}{3} < \frac{n}{9}$. Because n is an integer, there are no integers between $\frac{n}{9} - \frac{1}{9}$ and $\frac{n}{9}$, so $\lfloor \frac{x}{3} \rfloor$ is constant; similarly, $\lceil 3x \rceil$ is constant on the same intervals. Let l_n be the value of f on the interval I_n, and let $L_n = f(\frac{n}{3})$, the value at the right end of the interval I_n. If n is not a multiple of 9, then $l_n = L_n$, because as x increases from $n - \varepsilon$ to n, the floor function does not increase. This means that f is actually constant on the half-closed interval $(\frac{n-1}{3}, \frac{n}{3}]$. If neither n nor $n+1$ are multiples of 9, then $l_{n+1} = l_n + 1$. However if n is a multiple of 9, then $L_n = l_n + 1$ and $l_{n+1} = L_n + 1$. (The value of $f(x)$ increases when x increases from $n - \varepsilon$ to n, as well as going from n to $n + \varepsilon$.)

Hence on each interval of the form $(3n-3, 3n)$, the graph of f looks like 9 steps of height 1 and width $\frac{1}{3}$, all open on the left and closed on the right except for the last step, which is open on both ends. Between the intervals $(3n-3, 3n)$ and $(3n, 3n+3)$, $f(x)$ increases by 2, with $f(3n)$ halfway between steps. This graph is shown below:



On each interval $(3n-3, 3n)$, the average rate of change is $3 < \sqrt{11}$, so the steps move down relative $y = \sqrt{11}x$ within each interval. At the end of each interval, the graph of f rises relative to $y = \sqrt{11}x$. Thus the last intersection point between $f(x)$ and $\sqrt{11}x$ will be on the ninth step of one of these intervals. Suppose this intersection point lies in the interval $(3k-3, 3k)$. The ninth step is of height $10k-1$. Set $x = 3k - r$, where $r < \frac{1}{3}$. Then the solution is the largest k for which

$$\begin{aligned} 10k - 1 &= \sqrt{11}(3k - r) \qquad \left(0 < r < \frac{1}{3}\right) \\ k(10 - 3\sqrt{11}) &= 1 - \sqrt{11}r < 1 \\ k &< \frac{1}{10 - 3\sqrt{11}} = 10 + 3\sqrt{11} < 20. \end{aligned}$$

Because $0 < 19(10 - 3\sqrt{11}) < 1$, $k = 19$ implies a value of r between 0 and $\frac{1}{\sqrt{11}}$. And because $\frac{1}{\sqrt{11}} < \frac{1}{3}$,

$$x = 3k - r = \frac{10k - 1}{\sqrt{11}} = \boxed{\frac{189\sqrt{11}}{11}}$$

is the largest solution to $f(x) = \sqrt{11}x$.

Alternate Solution: Let x be the largest real number for which $\left\lfloor \frac{x}{3} \right\rfloor + \lceil 3x \rceil = \sqrt{11}x$. Because the left-hand side of this equation is an integer, it is simpler to work with $n = \sqrt{11}x$ instead of x. The equation becomes

$$\left\lfloor \frac{n}{3\sqrt{11}} \right\rfloor + \left\lceil \frac{3n}{\sqrt{11}} \right\rceil = n.$$

A little bit of computation shows that $\frac{1}{3\sqrt{11}} + \frac{3}{\sqrt{11}} > 1$, so the equation cannot hold for large values of n. To make this explicit, write

$$\left\lfloor \frac{n}{3\sqrt{11}} \right\rfloor = \frac{n}{3\sqrt{11}} - r \quad \text{and} \quad \left\lceil \frac{3n}{\sqrt{11}} \right\rceil = \frac{3n}{\sqrt{11}} + s,$$

where r and s are real numbers between 0 and 1. (If $n \neq 0$, then r and s are strictly between 0 and 1.) Then

$$\begin{aligned} 1 > r - s &= \Big(\frac{n}{3\sqrt{11}} - \left\lfloor \frac{n}{3\sqrt{11}} \right\rfloor\Big) - \Big(\left\lceil \frac{3n}{\sqrt{11}} \right\rceil - \frac{3n}{\sqrt{11}}\Big) \\ &= \Big(\frac{n}{3\sqrt{11}} + \frac{3n}{\sqrt{11}}\Big) - \Big(\left\lfloor \frac{n}{3\sqrt{11}} \right\rfloor + \left\lceil \frac{3n}{\sqrt{11}} \right\rceil\Big) \\ &= n\Big(\frac{1}{3\sqrt{11}} + \frac{3}{\sqrt{11}} - 1\Big), \end{aligned}$$

so $n < 1/\big(\frac{1}{3\sqrt{11}} + \frac{3}{\sqrt{11}} - 1\big) = 99 + 30\sqrt{11} = 198.45\ldots$.

Use trial and error with $n = 198, 197, 196, \ldots$, to find the value of n that works. Computing the first row of the following table to three decimal digits, and computing both $\frac{1}{3\sqrt{11}}$ and $\frac{3}{\sqrt{11}}$ to the same degree of accuracy, allows one to calculate the remaining rows with acceptable round-off errors.

n	$n/(3\sqrt{11})$	$3n/\sqrt{11}$
198	19.900	179.098
197	19.799	178.193
196	19.699	177.289
195	19.598	176.384
194	19.498	175.480
193	19.397	174.575
192	19.297	173.671
191	19.196	172.766
190	19.096	171.861
189	18.995	170.957

Because $n = 189 = 18 + 171$, the final answer is $x = \dfrac{\mathbf{189\sqrt{11}}}{\mathbf{11}}$.